

① Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function and let x^* be a solution to the equation $f(x^*) = 0$. We will approximate x^* using a rapidly converging sequence (x_n) , through Newton's method. At each iteration x_n , we consider the linear approximation $T_1(x)$ for $f(x)$ around x_n . We define the next iteration x_{n+1} by imposing the condition $T_1(x_{n+1}) = 0$.

a) Show that
$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

The first-order Taylor polynomial (tangent line) of f at $x = x_n$ is $T_1(x) = f(x_n) + f'(x_n)(x - x_n)$

According to Newton's method $\Rightarrow T_1(x_{n+1}) = 0$

Substitute $x = x_{n+1}$ into T_1

$$T_1(x_{n+1}) = f(x_n) + f'(x_n)(x_{n+1} - x_n)$$

$$0 = f(x_n) + f'(x_n)(x_{n+1} - x_n)$$

$$f'(x_n)(x_{n+1} - x_n) = -f(x_n)$$

$$x_{n+1} - x_n = -\frac{f(x_n)}{f'(x_n)}$$

$$x_{n+1} = -\frac{f(x_n)}{f'(x_n)} + x_n$$

$$\Rightarrow \text{Indeed, } x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

b) Show that if (x_n) is convergent, the $x_n \rightarrow x^*$.

Let $L \in \mathbb{R}$ st. $\lim_{n \rightarrow \infty} x_n = L$

Assume f' is continuous on a neighborhood of L and $f'(L) \neq 0$

iteration
 $\xrightarrow{\text{formula}}$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$\lim_{n \rightarrow \infty} x_{n+1} = \lim_{n \rightarrow \infty} \left(x_n - \frac{f(x_n)}{f'(x_n)} \right)$$

f and f' are continuous near $L \Rightarrow \lim_{n \rightarrow \infty} f(x_n) = f(L)$ and
 $\lim_{n \rightarrow \infty} f'(x_n) = f'(L)$

Substitute in the limit

$$\Rightarrow L = L - \frac{f(L)}{f'(L)}$$

$$0 = - \frac{f(L)}{\underbrace{f'(L)}_{\neq 0}} \Rightarrow f(L) = 0$$

$\Rightarrow L$ is a root of $f \Rightarrow L = x^*$

$\Rightarrow$ If (x_n) converges, then $x_n \rightarrow x^*$

d) Assume that x^* is a simple root to the equation $f(x^*) = 0$ and that f is twice continuously differentiable in a neighborhood of x^* . If x_0 is sufficiently close to x^* , prove that (x_n) converges quadratically to x^* , meaning that there exists $C > 0$ such that $e_{n+1} \leq C e_n^2$.

x^* is a simple root $\Rightarrow f'(x^*) \neq 0$

~~Assume~~ f twice continuously differentiable near x^*
 $\Rightarrow$ For x_n close enough to x^* :

1. $f'(x_n) \neq 0$, and $|f'(x_n)| \geq m > 0$
2. $|f''(x_n)| \leq M$ for some $M < \infty$.

c) says
$$e_{n+1} = \frac{f''(x_n)}{2f'(x_n)} e_n^2$$

$$\Rightarrow e_{n+1} \leq \frac{M}{2m} e_n^2$$

$$\text{Let } C = \frac{M}{2m} > 0$$

$$\Rightarrow e_{n+1} \leq C e_n^2$$

```

1  import numpy as np
2
3  def newton_sqrt(a, x0, max_iter=10): 1 usage
4      exact = np.sqrt(a)
5      x = x0
6      errors = []
7      for n in range(max_iter):
8          x_new = (x + a / x) / 2.0
9          error = abs(x_new - exact)
10         errors.append(error)
11         print(f"Iter {n}: x = {x_new:.15f}, error = {error:.2e}")
12         x = x_new
13     return errors
14
15     # Example: sqrt(2)
16     a = 2.0
17     x0 = 1.5
18     print(f"Computing sqrt({a}) with Newton's method, starting at x0 = {x0}\n")
19     errors = newton_sqrt(a, x0, max_iter: 6)
20

```

C:\Users\Crina\PycharmProjects\extra6\.venv\Scripts\python.exe C:\Users\Crina\PycharmProjects\extra6\extra6.py

Computing sqrt(2.0) with Newton's method, starting at x0 = 1.5

```

Iter 0: x = 1.4166666666666667, error = 2.45e-03
Iter 1: x = 1.414215686274510, error = 2.12e-06
Iter 2: x = 1.414213562374690, error = 1.59e-12
Iter 3: x = 1.414213562373095, error = 2.22e-16
Iter 4: x = 1.414213562373095, error = 2.22e-16
Iter 5: x = 1.414213562373095, error = 2.22e-16

```

Process finished with exit code 0